

Problem

Southwest Water, a renewable energy management company, owns and operates wind turbines across Devon. Due to recent unpredictable weather conditions and reactive plant operations, the Den Brook Wind Farm has been operating at a financial loss. In this project, I will create a data-driven solution to bring the plant back to profitability.

Constructing an application to achieve the following:

- **Minimise Operational Costs:** Build a short-term operational plan to optimise energy output and minimise manual adjustment errors.
- **Guide Long-Term Investments:** Evaluate historical operational data, employee absences, and weather pattern volatility to recommend long-term efficiency investments.

Application Workflow

- **Authentication:** Users log into the application using secure credentials (Username and LoginPassword).
- **Historical Analysis:** Users analyse WindData alongside WindPowerData to evaluate performance during volatile days, compare expected vs. actual energy outputs, and cross-reference operational gaps with employee absences.
- **Predictive Planning:** Users leverage PredictorModel timestamps and PredictionResult forecasts (wind speed, direction, and weather notes) to generate optimised blade angles and staffing plans.

Metric Definitions

- **Absences:** Stored in a single normalised table linked by UserID. Tracks WeekNumber (ex: Week 11 or 12), DaysAbsent, and AbsenceInformation (reason for absence).
- **Wind Data:** Recorded every 8 hours. Tracks WindSpeed in mph (INTEGER), WindDirection in degrees (INTEGER, where 0° = North, 90° = East), and WeatherInformation.
- **Wind Power Data:** Measures power generation in Watts (W). Tracks ExpectedEnergyProduction, ActualEnergyProduction, TurbineInformation (ex: Optimal/Suboptimal), ExpectedTurbineDirection, and ActualTurbineDirection. Shares the standard Timemarker timestamp with WindData.
- **Prediction Model & Results:** Tracks 12-hour forecast intervals (TimeOfPrediction) predicting WindSpeedPrediction (mph), WindDirectionPrediction (degrees), and ResultInformation.

Data Dictionary in 3NF

1. User Table

- UserID (PK, INTEGER, AUTOINCREMENT)
- Username (TEXT, UNIQUE, NOT NULL)
- LoginPassword (TEXT, NOT NULL)
- Sample Entry: 1 | john_doe | P@ssw0rd1 (10 total user rows)

2. Absences Table

- AbsenceID (PK, INTEGER, AUTOINCREMENT)
- UserID (FK, INTEGER, NOT NULL)
- WeekNumber (INTEGER, NOT NULL)
- DaysAbsent (TEXT, NOT NULL)
- AbsenceInformation (TEXT, NOT NULL)
- Sample Entry: 1 | 6 | 12 | 2024-03-20 | Maternity Leave

3. WindData Table

- WindDataID (PK, INTEGER, AUTOINCREMENT)
- Timemarker (TEXT, UNIQUE, NOT NULL)
- WindSpeed (INTEGER, NOT NULL)
- WeatherInformation (TEXT, NOT NULL)
- WindDirection (INTEGER, NOT NULL)
- Sample Entry: 1 | 2024-03-13 08:00:00 | 14 | Sunny | 0 (30 total rows)

4. WindPowerData Table

- WindPowerDataID (PK, INTEGER, AUTOINCREMENT)
- Timemarker (FK, TEXT, NOT NULL)
- ExpectedEnergyProduction (INTEGER, NOT NULL)
- ActualEnergyProduction (INTEGER, NOT NULL)
- TurbineInformation (TEXT, NOT NULL)
- ExpectedTurbineDirection (INTEGER, NOT NULL)
- ActualTurbineDirection (INTEGER, NOT NULL)
- Sample Entry: 1 | 2024-03-13 08:00:00 | 280 | 280 | Optimal condition | 0 | 0 (30 total rows)

5. PredictorModel Table

- PredictorModelID (PK, INTEGER, AUTOINCREMENT)
- PredictorInformation (TEXT, NOT NULL)
- TimeOfPrediction (TEXT, UNIQUE, NOT NULL)
- Sample Entry: 1 | Future Analysis | 2024-03-23 00:00:00 (20 total rows)

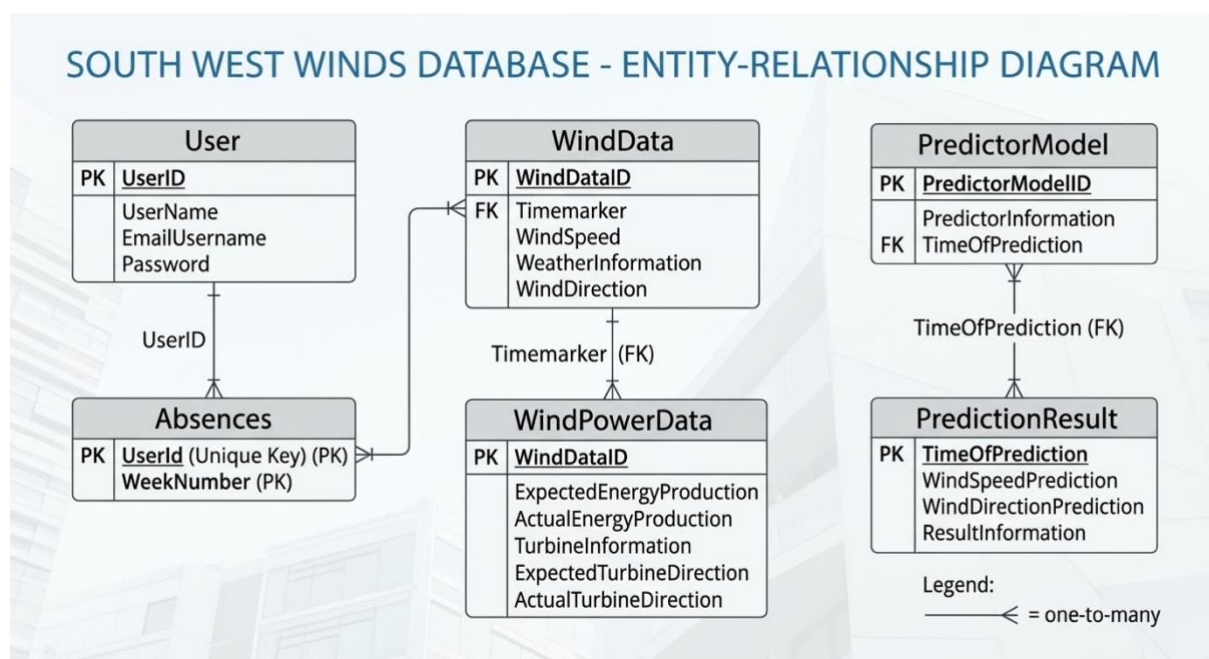
6. PredictionResult Table

- PredictionResultID (PK, INTEGER, AUTOINCREMENT)
- TimeOfPrediction (FK, TEXT, NOT NULL)
- WindSpeedPrediction (INTEGER, NOT NULL)
- WindDirectionPrediction (INTEGER, NOT NULL)
- ResultInformation (TEXT, NOT NULL)
- Sample Entry: 1 | 2024-03-23 00:00:00 | 10 | 0 | Clear (20 total rows)

Normalisation Proof

- 1NF: All column values are atomic (ex: integer degrees instead of degree strings), with no multi-valued attributes or repeating groups.
- 2NF: All non-key fields depend entirely on single-column primary keys.
- 3NF: Eliminates transitive dependencies by replacing duplicate weekly tables with a unified Absences entity linked via foreign keys (UserID). Validated all date strings to adhere strictly to standardised calendar limits.

ERD



Analysis

Historical Analysis Key Findings

- **Optimal Operations:** Peak recorded wind speed of 30 mph produced 600 W when the turbine was aligned with the wind direction (0°) and in optimal condition. Minimum wind speed of 5 mph generated 100 W.
- **Direction Misalignment:** On 2024-03-20 at 16:00:00, wind direction shifted to 55°. The turbine direction was incorrectly set to 80°, dropping power output from an expected 500 W to an actual 425 W.
- **Suboptimal Maintenance:** On 2024-03-21 at 08:00:00, mechanical degradation cut expected output in half from 320 W down to 160 W due to delayed maintenance.
- **Correlation:** A strong positive correlation exists between wind speed and energy output (approximately 1 mph $\approx$ 20 W generated). Mean wind speed across sample records was calculated via Python (numpy) as 15.13 mph, translating to an expected baseline output of 302.6 W.
- **Root Cause of Operational Losses:** Misalignments coincided directly with staff shortages recorded in the Absences table. Laura Miller's extended absence (maternity leave) combined with short-term sick leave during volatile weather days (changes > 10 mph or > 60° direction shift) left remaining operators unable to handle manual recalibrations.

Long-Term Recommendations

- **Strategic Staffing:** Recruit replacement operators to cover extended leave (ex:Laura Miller's maternity leave) to ensure adequate coverage during shifts with rapid weather shifts.
- **Targeted Innovation:** Invest in real-time automated yaw-control systems and predictive health monitoring sensors. Automating direction adjustments directly eliminates manual operator errors during volatile shifts and reduces ongoing labor overhead.
- **Dynamic Power Threshold Alert:** Configure the monitoring software to flag any shift where actual output falls >15% below the 302.6 W baseline during standard 15 mph wind conditions. This gives on-duty staff a real-time signal to adjust manual blade angles immediately rather than discovering lost yield at the end of a shift.
- **Capacity Planning & Contract Guarantees:** Use the 302.6 W baseline average to establish realistic minimum power-purchase agreement (PPA) supply commitments with local Devon energy distributors. Anchoring supply contracts to this statistically derived baseline prevents over-promising energy generation and avoids financial penalties during lower-wind months.

Short-Term Recommendations

Date	Forecasted Weather	Direction Recommendation	Staffing Action
2024-03-23	Clear	Set turbine to 0° at 00:00 and 0° at 12:00	Standard staffing; mild conditions.
2024-03-24	Sunny	Set turbine to 0° at 00:00 and 0° at 12:00	Standard staffing; stable conditions.
2024-03-25	Thunderstorm / High Volatility	Set turbine to 45° at 00:00 and 60° at 12:00	Increase staffing. Rapid wind jump (3 to 25 mph); active monitoring required.
2024-03-26	Thunderstorm / Shift	Set turbine to 60° at 00:00 and 60° at 12:00	Maintain elevated staffing to handle sudden velocity shifts.
2024-03-27	Sunny / Clear	Set turbine to 60° at 00:00 and 60° at 12:00	Standard staffing.
2024-03-28	Light Rain / Cloudy	Set turbine to 45° at 00:00 and 45° at 12:00	Standard staffing.
2024-03-29	Clear / Cloudy	Set turbine to 40° at 00:00 and 40° at 12:00	Standard staffing.
2024-03-30	Rain / Major Direction Shift	Set turbine to 20° at 00:00 and 200° at 12:00	Increase staffing. 180° shift following heavy rain requires active monitoring.
2024-03-31	Light Rain / Shift	Set turbine to 200° at 00:00 and 180° at 12:00	Maintain elevated staffing to oversee directional adjustments.

Reflection from Workshops at University

- Conceptualisation: Explored business domain options (inventory management, financial portfolio, renewable energy). Selected a renewable energy management approach combining cost-reduction strategies with human resource monitoring.
- Database Design & ERD: Received peer feedback on initial entity relationships. Applied functional dependency analysis to refine unnormalised tables into properly structured 1NF, 2NF, and 3NF designs.

- Implementation & Analytics: Finalised business logic, schema design, and analytical queries. Implemented relational schema into SQLite and utilised Python (pyodbc, numpy) inside Miniconda environments to extract raw data and compute mean wind metrics for baseline power modeling.